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Roll No:-2401038
AI and DS
Division:T1
Batch:T12
Subject: ASD & DevOps

EXPERIMENT:-2

Aim: To Containerize Application using Docker

This experiment maps to LO4.

LO4: Containerisation, orchestrate and deploy applications using Docker, Compose, Kubernetes and cloud infrastructure tools.

Theory:

Docker is an open-source containerization platform used to develop, package, and run applications in isolated environments called containers. It allows developers to package an application along with all its required dependencies, libraries, and configuration files into a single unit called a Docker image. These images can be shared and executed consistently across different systems, making application deployment faster and more reliable. Docker uses operating-system-level virtualization, which makes it lightweight and more efficient compared to traditional virtual machines.

Containerization is a technology used to package an application and all its dependencies into a lightweight, isolated execution environment called a container. A container includes the application code, runtime, libraries, and configuration files required for the application to run. Containers provide consistency across different environments because the application behaves the same regardless of where it is deployed. Containerization improves portability, resource utilization, scalability, and simplifies application development and deployment.

Docker follows a client-server architecture consisting of the Docker Client, Docker Engine, Docker Images, and Docker Containers. The Docker Client provides a command-line interface through which users execute Docker commands. The Docker Engine is responsible for creating, managing, and running containers. Docker Images act as templates containing the application and its dependencies, while Docker Containers are the running instances created from these images.

To install Docker on Windows, first enable hardware virtualization from BIOS/UEFI settings. Then enable required Windows features such as Windows Subsystem for Linux (WSL), Virtual Machine Platform, and Hyper-V (if supported). Install and configure WSL 2 using the command wsl --install. After that, download and install Docker Desktop for Windows and select the WSL 2

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based engine option during installation. Once Docker Desktop is installed and running, verify the installation using commands such as docker --version and docker info.

Term	Description
Docker Image	A blueprint or template used to create containers.
Docker Container	A running instance of a Docker image.

Table 2: Docker Commands

Command	Explanation
docker --version	Displays the installed Docker version.
docker info	Shows detailed information about Docker Engine and system configuration.
docker search ubuntu	Searches Docker Hub for available Ubuntu images.
docker pull ubuntu	Downloads the Ubuntu image from Docker Hub to the local system.
docker images	Displays all Docker images available on the system.
docker run --name mycontainer ubuntu	Creates and runs a container named mycontainer using the Ubuntu image.
docker run -it ubuntu bash	Runs an Ubuntu container interactively and opens a bash shell inside it.
docker ps	Displays currently running containers.
docker ps -a	Displays all containers including stopped containers.
docker start mycontainer	Starts an existing stopped container.
docker stop mycontainer	Stops a running container.

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docker restart mycontainer	Restarts a container.
docker exec -it mycontainer bash	Opens a command-line shell inside a running container.
docker logs mycontainer	Displays the logs generated by a container.
docker inspect mycontainer	Displays detailed information about a container.
docker stats	Shows CPU and memory usage statistics of running containers.
docker rm mycontainer	Removes a container from the system.
docker rmi ubuntu	Removes a Docker image from the system.
docker pull nginx	Downloads the Nginx image from Docker Hub.
docker run -d --name webserver -p 8080:80 nginx	Runs an Nginx container in detached mode and maps port 8080 of the host to port 80 of the container.
docker stop webserver	Stops the Nginx container.
docker rm webserver	Removes the Nginx container.
docker image prune	Removes unused Docker images.
docker container prune	Removes stopped containers.

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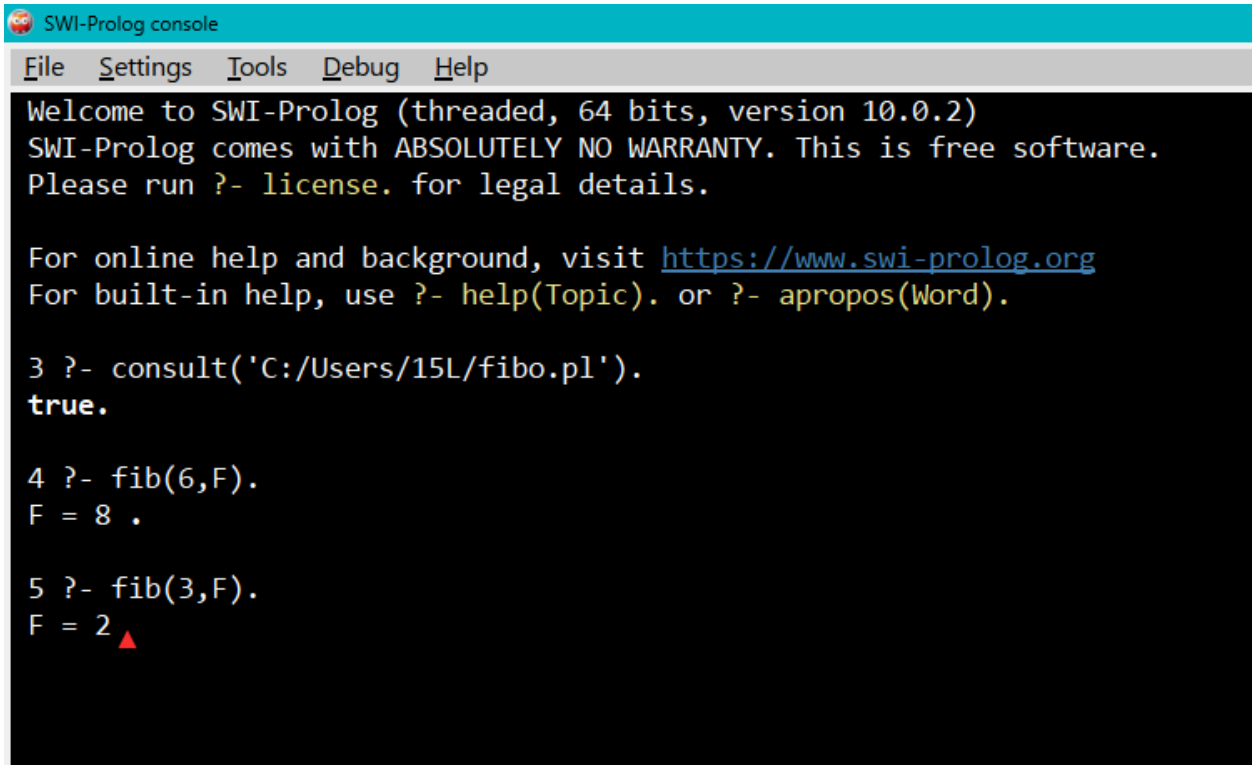
Output:

```
C:\Users\15L>docker  version
Client:
 Version:           29.2.1
 API version:       1.53
 Go version:        go1.25.6
 Git commit:        a5c7197
 Built:             Mon Feb  2 17:20:16 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.63.0 (220185)
 Engine:
  Version:          29.2.1
  API version:      1.53 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       6bc6209
  Built:            Mon Feb  2 17:17:24 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

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```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```



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SWI-Prolog console

File Settings Tools Debug Help

```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

12 ?- consult('C:/Users/15L/greatest.pl').
true.

13 ?- greatest(10, 14, 50, G).
G = 50.

14 ?- greatest(15, 140, 101, G).
G = 140
```

```
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

C:\Users\15L> docker search ubuntu
NAME                                DESCRIPTION                                     STARS     OFFICIAL
ubuntu                              Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                        Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                        BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                        Apache Kafka, a distributed event streaming …   62
ubuntu/apache2                      Apache, a secure & extensible open-source HT…   104
ubuntu/zookeeper                    ZooKeeper maintains configuration informatio…   15
ubuntu/prometheus                   Prometheus is a systems and service monitori…   80
ubuntu/jre                          Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet                Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                       A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                        MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                     PostgreSQL is an open source object-relation…   45
ubuntu/redis                        Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime               Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                  Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                      Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                    Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                          Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                     Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                        Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                         Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                      Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/alertmanager                 Ubuntu Rock for Alertmanager. It handles ale…   1
ubuntu/opentelemetry-collector      This is a rock for the OpenTelemetry Collect…   0

C:\Users\15L>
```

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```
C:\Users\15L>docker --help
Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  bake        Build from a file
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Authenticate to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  agent*      create or run AI agents
  ai*         Docker AI Agent - Ask Gordon
  builder     Manage builds
  buildx*     Docker Buildx
  compose*    Docker Compose
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container
  desktop*    Docker Desktop commands
  extension*  Manages Docker extensions
  image       Manage images
  init*       Creates Docker-related starter files for your project
  manifest    Manage Docker image manifests and manifest lists
  mcp*        Docker MCP Plugin
  model*      Docker Model Runner
  network     Manage networks
  offload*    Docker Offload
  plugin      Manage plugins
  sandbox*    Docker Sandbox
  sbom*       View the packaged-based Software Bill Of Materials (SBOM) for an image
  scout*      Docker Scout
  system      Manage Docker
  volume      Manage volumes

Swarm Commands:
  swarm       Manage Swarm
```

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```
Commands:
  attach      Attach local standard input, output, and error streams to a running co
  commit      Create a new image from a container's changes
  cp          Copy files/folders between a container and the local filesystem
  create      Create a new container
  diff        Inspect changes to files or directories on a container's filesystem
  events      Get real time events from the server
  export      Export a container's filesystem as a tar archive
  history     Show the history of an image
  import      Import the contents from a tarball to create a filesystem image
  inspect     Return low-level information on Docker objects
  kill        Kill one or more running containers
  load        Load an image from a tar archive or STDIN
  logs        Fetch the logs of a container
  pause       Pause all processes within one or more containers
  port        List port mappings or a specific mapping for the container
  rename      Rename a container
  restart     Restart one or more containers
  rm          Remove one or more containers
  rmi         Remove one or more images
  save        Save one or more images to a tar archive (streamed to STDOUT by defaul
  start       Start one or more stopped containers
  stats       Display a live stream of container(s) resource usage statistics
  stop        Stop one or more running containers
  tag         Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE
  top         Display the running processes of a container
  unpause     Unpause all processes within one or more containers
  update      Update configuration of one or more containers
  wait        Block until one or more containers stop, then print their exit codes

Global Options:
      --config string      Location of client config files (default
                           "C:\\Users\\15L\\.docker")
  -c, --context string     Name of the context to use to connect to the
                           daemon (overrides DOCKER_HOST env var and
                           default context set with "docker context use")
  -D, --debug              Enable debug mode
  -H, --host string        Daemon socket to connect to
  -l, --log-level string   Set the logging level ("debug", "info",
                           "warn", "error", "fatal") (default "info")
      --tls                Use TLS; implied by --tlsverify
      --tlscacert string   Trust certs signed only by this CA (default
                           "C:\\Users\\15L\\.docker\\ca.pem")
      --tlscert string     Path to TLS certificate file (default
                           "C:\\Users\\15L\\.docker\\cert.pem")
      --tlskey string      Path to TLS key file (default
                           "C:\\Users\\15L\\.docker\\key.pem")
      --tlsverify          Use TLS and verify the remote
  -v, --version            Print version information and quit
```

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```
C:\Users\15L>docker run --name mycontainer ubuntu

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                          PORTS      NAMES
c766f6ae06d0   ubuntu         "/bin/bash"              10 seconds ago   Exited (0) 10 seconds ago                  mycontainer
656823eb03e6   ubuntu         "bash"                   4 minutes ago    Exited (127) About a minute ago            goofy_taussig
e78acc6291ff   redis          "docker-entrypoint.s…"   19 hours ago     Exited (255) 11 minutes ago     6379/tcp   my-redis-db
503c4b95926e   ubuntu         "/bin/bash"              19 hours ago     Exited (127) 19 hours ago                  hungry_ardinghelli
b5c4481e06b8   ubuntu         "/bin/bash"              19 hours ago     Exited (0) 19 hours ago                    eloquent_wing
55d70b7aedf0   ubuntu         "/bin /bash"             19 hours ago     Created                                    crazy_carver
ac0e674ab8d5   redis          "docker-entrypoint.s…"   3 months ago     Exited (0) 3 months ago                    romantic_swanson
44881704ad11   1c054d54ecd1   "docker-entrypoint.s…"   4 months ago     Exited (0) 4 months ago                    elated_jennings
4decb73d97e3   1c054d54ecd1   "docker-entrypoint.s…"   4 months ago     Exited (255) 4 months ago       6379/tcp   inspiring_ritchie
61dcc809f9a9   1c054d54ecd1   "docker-entrypoint.s…"   4 months ago     Exited (255) 4 months ago       6379/tcp   magical_dubinsky

C:\Users\15L>docker start mycontainer
mycontainer

C:\Users\15L>docker stop mycontainer
mycontainer

C:\Users\15L>docker start mycontainer
mycontainer
```

```
C:\Users\15L>mkdir aids

C:\Users\15L>cd aids

C:\Users\15L\aids>echo "Hello Docker Container" > file.txt
```

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                          PORTS      NAMES
c766f6ae06d0   ubuntu         "/bin/bash"              6 minutes ago    Exited (0) 5 minutes ago                   myconta
iner
656823eb03e6   ubuntu         "bash"                   10 minutes ago   Exited (127) 7 minutes ago                 goofy_t
aussig
e78acc6291ff   redis          "docker-entrypoint.s…"   19 hours ago     Exited (255) 17 minutes ago     6379/tcp   my-redi
s-db
503c4b95926e   ubuntu         "/bin/bash"              19 hours ago     Exited (127) 19 hours ago                  hungry_
ardinghelli
b5c4481e06b8   ubuntu         "/bin/bash"              19 hours ago     Exited (0) 19 hours ago                    eloquen
t_wing
55d70b7aedf0   ubuntu         "/bin /bash"             19 hours ago     Created                                    crazy_c
arver
ac0e674ab8d5   redis          "docker-entrypoint.s…"   3 months ago     Exited (0) 3 months ago                    romanti
c_swanson
44881704ad11   1c054d54ecd1   "docker-entrypoint.s…"   4 months ago     Exited (0) 4 months ago                    elated_
jennings
4decb73d97e3   1c054d54ecd1   "docker-entrypoint.s…"   4 months ago     Exited (255) 4 months ago       6379/tcp   inspiri
ng_ritchie
61dcc809f9a9   1c054d54ecd1   "docker-entrypoint.s…"   4 months ago     Exited (255) 4 months ago       6379/tcp   magical
_dubinsky
```

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```
C:\Users\15L>docker inspect mycontainer
[
    {
        "Id": "c766f6ae06d0d85f2a5a8ea94d69e619c08b43f867b37c702a7eef85e4357bfc",
        "Created": "2026-07-21T05:07:41.343159392Z",
        "Path": "/bin/bash",
        "Args": [],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-21T05:08:43.572535357Z",
            "FinishedAt": "2026-07-21T05:08:43.694738032Z"
        },
        "Image": "sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb",
        "ResolvConfPath": "/var/lib/docker/containers/c766f6ae06d0d85f2a5a8ea94d69e619c08b43f867b37c702a7eef85e4357bfc/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/c766f6ae06d0d85f2a5a8ea94d69e619c08b43f867b37c702a7eef85e4357bfc/hostname",
        "HostsPath": "/var/lib/docker/containers/c766f6ae06d0d85f2a5a8ea94d69e619c08b43f867b37c702a7eef85e4357bfc/hosts",
        "LogPath": "/var/lib/docker/containers/c766f6ae06d0d85f2a5a8ea94d69e619c08b43f867b37c702a7eef85e4357bfc/c766f6ae06d0d85f2a5a8ea94d69e619c08b43f867b37c702a7eef85e4357bfc-json.log",
        "Name": "/mycontainer",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                51,
```

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```
C:\Users\15L>docker stats --no-stream

C:\Users\15L>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
d26f27cc8c41: Pull complete
82454cdbf456: Pull complete
3c7ab7949321: Pull complete
cacfcdd01f30: Pull complete
2bedaf25031a: Pull complete
062e450697fa: Pull complete
b6698f04e005: Pull complete
6c496f5b5050: Download complete
ea1d76ccc2c6: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

```
C:\Users\15L>docker logs webserver

C:\Users\15L>docker stop webserver
webserver

C:\Users\15L>docker rm webserver
webserver

C:\Users\15L>$docker rmi ubuntu
'$docker' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\15L>docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] docker info

C:\Users\15L>docker images

IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
my-app:1.0         07c9f9065fb5       157MB         41.6MB
my-app:latest      e1b8c6a9dbf7       157MB         41.6MB
my-ubuntu-app:v1   9ddf795e7b17       157MB         41.6MB
nginx:latest       5a88c9c45479       241MB           66MB
redis:latest       009cc37796fb       204MB         55.3MB    U
ubuntu:latest      3131b4cc82a7       161MB         45.3MB    U
```

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Conclusion: I learned how to containerize an application using Docker by creating a Docker image and running it as a container. I understood the purpose of Dockerfiles, images, and containers, and learned how Docker provides a consistent execution environment across different systems. This experiment also helped me understand the basics of application deployment, portability, and container management using Docker commands.